HW0505

2026-05-19

```
# install.packages("car")
# install.packages("AER")
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
library(car)  # 解決 linearHypothesis 的錯誤
```

```
## Warning: package 'car' was built under R version 4.5.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.5.3
```

```
library(AER)  # 預防後面 ivreg 出現錯誤
```

```
## Warning: package 'AER' was built under R version 4.5.3
```

```
## Loading required package: lmtest
```

```
## Warning: package 'lmtest' was built under R version 4.5.3
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.5.3
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
## Loading required package: sandwich
```

```
## Warning: package 'sandwich' was built under R version 4.5.3
```

```
## Loading required package: survival
```

```
# 11.24
rm(list=ls())
library(readxl)
fultonfish <- read.csv("C:/Users/tiffa/OneDrive/Desktop/eco_fin/fultonfish.csv")
data('fultonfish')
```

```
## Warning in data("fultonfish"): data set 'fultonfish' not found
```

```
# a.
rf_price = lm(lprice ~ mon + tue + wed + thu + stormy + mixed, data = fultonfish)
cat("ANS of a.", "\n")
```

```
## ANS of a.
```

```
summary(rf_price)
```

```
##
## Call:
## lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
##     data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92281 -0.18819  0.00813  0.22404  0.68381
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
## mon         -0.10825    0.10339  -1.047  0.29753
## tue         -0.06609    0.10103  -0.654  0.51446
## wed         -0.04927    0.10377  -0.475  0.63591
## thu          0.03935    0.10071   0.391  0.69681
## stormy       0.44634    0.07921   5.635 1.51e-07 ***
## mixed        0.23688    0.07851   3.017  0.00321 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3413 on 104 degrees of freedom
## Multiple R-squared:  0.245,  Adjusted R-squared:  0.2014
## F-statistic: 5.624 on 6 and 104 DF,  p-value: 4.349e-05
```

```
linearHypothesis(rf_price, c("stormy = 0", "mixed = 0"))
```

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq     F    Pr(>F)
## 1    106 15.876
## 2    104 12.115  2    3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("STORMY係數為0.44634。且p-val為顯著。在暴風雨下，魚價會顯著上升。", "\n")
```

```
## STORMY係數為0.44634。且p-val為顯著。在暴風雨下，魚價會顯著上升。
```

```
cat("MIXED係數為0.23688。且p-val為顯著。在暴風雨下，混合天啟會導致魚價上升。", "\n")
```

```
## MIXED係數為0.23688。且p-val為顯著。在暴風雨下，混合天啟會導致魚價上升。
```

```
cat("Joint Significant Test 確認上述兩個變數是否為有效變數。因為p-val小於0.05，所以拒絕虛無假設。天氣變
數對內生變數有顯著解釋力", "\n")
```

```
## Joint Significant Test 確認上述兩個變數是否為有效變數。因為p-val小於0.05，所以拒絕虛無假設。天氣變
## 數對內生變數有顯著解釋力
```

```
# b.
demand_iv = ivreg(lquan ~ lprice + mon + tue + wed + thu |
                     mon + tue + wed + thu + stormy + mixed, data = fultonfish)
cat("ANS of b.", "\n")
```

```
## ANS of b.
```

```
summary(demand_iv)
```

```
##
## Call:
## ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
##     tue + wed + thu + stormy + mixed, data = fultonfish)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
## lprice      -0.93014    0.35340  -2.632  0.00977 **
## mon         -0.01190    0.20837  -0.057  0.95456
## tue         -0.52583    0.20230  -2.599  0.01068 *
## wed         -0.56262    0.20696  -2.719  0.00767 **
## thu          0.09987    0.20285   0.492  0.62350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared: 0.185,   Adjusted R-squared: 0.1462
## Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

```
cat("Lquan hat = 8.54023-0.93014*lprice-0.0119*mom-0.52583*tue-0.56262*wed+0.009987*thu", "\n")
```

```
## Lquan hat = 8.54023-0.93014*lprice-0.0119*mom-0.52583*tue-0.56262*wed+0.009987*thu
```

```
cat("lprice係數增加，但同為負號，且p值下降小於0.01，落在1%的reject region內。而table 11.5的lprice係數只能拒絕5%的虛無假設", "\n")
```

```
## lprice係數增加，但同為負號，且p值下降小於0.01，落在1%的reject region內。而table 11.5的lprice係數只能拒絕5%的虛無假設
```

```
# c.
cat("ANS of c.", "\n")
```

```
## ANS of c.
```

```
summary(demand_iv, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
##     tue + wed + thu + stormy + mixed, data = fultonfish)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
## lprice      -0.93014    0.35340  -2.632  0.00977 **
## mon         -0.01190    0.20837  -0.057  0.95456
## tue         -0.52583    0.20230  -2.599  0.01068 *
## wed         -0.56262    0.20696  -2.719  0.00767 **
## thu          0.09987    0.20285   0.492  0.62350
##
## Diagnostic tests:
##                   df1 df2 statistic  p-value
## Weak instruments   2 104    16.140 7.86e-07 ***
## Wu-Hausman         1 104     1.489    0.225
## Sargan             1  NA     0.772    0.380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared: 0.185,   Adjusted R-squared: 0.1462
## Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

```
cat("Sargan test要檢定工具變數與需求函數誤差像無關。表中p-val=0.38 can not reject H0。代表所選的工具變數具備外生性，模性設定是穩健的。", "\n")
```

```
## Sargan test要檢定工具變數與需求函數誤差像無關。表中p-val=0.38 can not reject H0。代表所選的工具變數具備外生性，模性設定是穩健的。
```

```
# d.
cat("ANS of d.", "\n")
```

```
## ANS of d.
```

```
linearHypothesis(rf_price, c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))
```

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    108 12.408
## 2    104 12.115  4   0.29256 0.6279 0.6437
```

```
cat("ANS of d.", "\n")
```

```
## ANS of d.
```

```
cat("H0：mon=tue=wed=thu=0 vs. H1：at least one is not 0", "\n")
```

```
## H0：mon=tue=wed=thu=0 vs. H1：at least one is not 0
```

```
cat("p-val=0.6437 > 0.05。not reject H0。代表在恐至天氣因素後，星期一至四對於魚的價格無顯著聯合影響，即市場價格波動來自於天氣所影響的供給面變動，並非每週固定週期的需求供給變動。", "\n")
```

```
## p-val=0.6437 > 0.05。not reject H0。代表在恐至天氣因素後，星期一至四對於魚的價格無顯著聯合影響，即市場價格波動來自於天氣所影響的供給面變動，並非每週固定週期的需求供給變動。
```

```
cat("此題詞無法估計供給函數，因為星期一至四的聯合顯著性檢定無法reject H0。若要估計函數，至捨需要一個只能影響需求但不能影響供給的工具變數，但今天微星期幾此一項變數不符合，因此無法透過需求曲線移動勾勒出供給曲線", "\n")
```

```
## 此題詞無法估計供給函數，因為星期一至四的聯合顯著性檢定無法reject H0。若要估計函數，至捨需要一個只能影響需求但不能影響供給的工具變數，但今天微星期幾此一項變數不符合，因此無法透過需求曲線移動勾勒出供給曲線
```

```
# 22.28
truffles <- read.csv("C:/Users/tiffa/OneDrive/Desktop/eco_fin/truffles.csv")
# b.
library(AER)
data("truffles")
```

```
## Warning in data("truffles"): data set 'truffles' not found
```

```
# 將價格 P 作為被解變數，使用 Q, PS, DI 為需求方程的解釋變數
demand_2sls <- ivreg(p ~ q + ps + di | ps + di + pf, data = truffles)

# 供給方程：價格 P 被解出，Q 和 PF 作為解釋變數
supply_2sls <- ivreg(p ~ q + pf | di + ps + pf, data = truffles)

cat("ANS of b.", "\n")
```

```
## ANS of b.
```

```
summary(demand_2sls, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -39.661  -6.781   2.410   8.320  20.251
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -11.428     13.592  -0.841  0.40810
## q             -2.671      1.175  -2.273  0.03154 *
## ps             3.461      1.116   3.103  0.00458 **
## di            13.390      2.747   4.875 4.68e-05 ***
##
## Diagnostic tests:
##                  df1 df2 statistic  p-value
## Weak instruments   1  26     17.48 0.000291 ***
## Wu-Hausman         1  25    120.03 4.92e-11 ***
## Sargan             0  NA        NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.17 on 26 degrees of freedom
## Multiple R-Squared: 0.5567,  Adjusted R-squared: 0.5056
## Wald test: 17.37 on 3 and 26 DF,  p-value: 2.137e-06
```

```
summary(supply_2sls, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = p ~ q + pf | di + ps + pf, data = truffles)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.7983 -2.3440 -0.6281  2.4350 11.1600
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
## q             2.9367     0.2158   13.61 1.32e-13 ***
## pf            2.9585     0.1560   18.97  < 2e-16 ***
##
## Diagnostic tests:
##                  df1 df2 statistic  p-value
## Weak instruments   2  26    28.934 2.44e-07 ***
## Wu-Hausman         1  26     7.051   0.0134 *
## Sargan             1  NA     1.544   0.2140
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.399 on 27 degrees of freedom
## Multiple R-Squared: 0.9486,  Adjusted R-squared: 0.9448
## Wald test: 232.7 on 2 and 27 DF,  p-value: < 2.2e-16
```

```r
# c.
# 取得需求方程式係數
coefs <- coef(demand_2sls)

# 計算自變數平均值
means <- colMeans(truffles[, c("q", "ps", "di")])

# 在平均值下估計價格 P
P_at_means <- coefs["(Intercept)"] +
              coefs["q"] * means["q"] +
              coefs["ps"] * means["ps"] +
              coefs["di"] * means["di"]
cat("ANS of c.", "\n")
```

```
## ANS of c.
```

```r
cat("平均價格 P =", P_at_means, "\n")
```

```
## 平均價格 P = 62.724
```

```r
cat("平均數量 Q =", means["q"], "\n")
```

```
## 平均數量 Q = 18.45833
```

```r
# 計算需求價格彈性
delta_2 <- coefs["q"]
elasticity <- (1 / delta_2) * (P_at_means / means["q"])
cat("需求價格彈性 =", elasticity, "\n")
```

```
## 需求價格彈性 = -1.272464
```

```r
# d.
library(ggplot2)
```

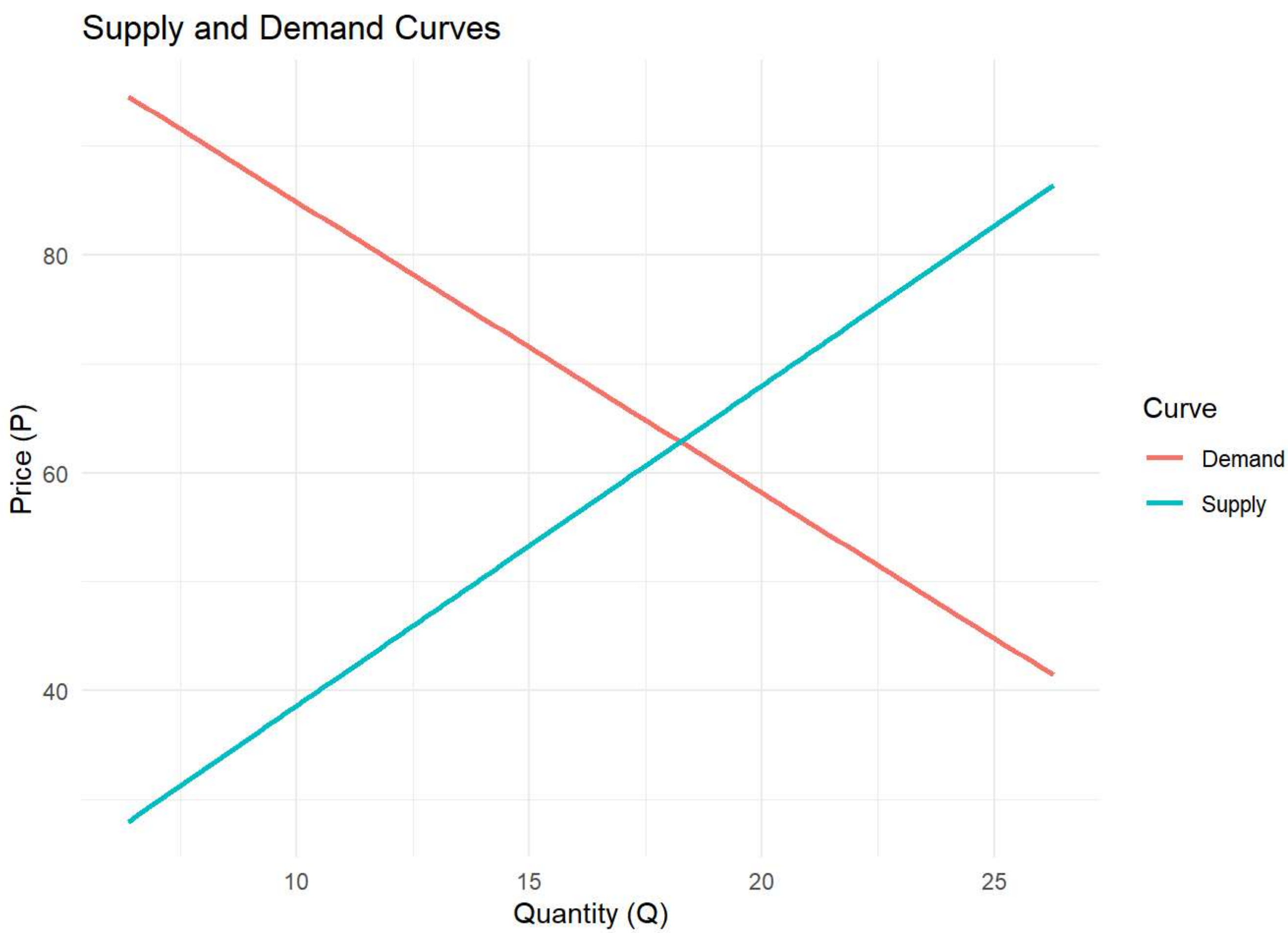
```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
# 需求方程式係數
d_coefs <- coef(demand_2sls)
# 供給方程式係數
s_coefs <- coef(supply_2sls)

# 固定外生變數
DI_star <- 3.5
PF_star <- 23
PS_star <- 22

# 設定 Q 範圍
Q_vals <- seq(min(truffles$q), max(truffles$q), length.out = 100)

# 計算需求價格
P_demand <- d_coefs["(Intercept)"] + d_coefs["q"] * Q_vals + d_coefs["ps"] * PS_star + d_coefs["di"] * DI_star

# 計算供給價格
P_supply <- s_coefs["(Intercept)"] + s_coefs["q"] * Q_vals + s_coefs["pf"] * PF_star

# 建立資料框
df <- data.frame(
  Q = rep(Q_vals, 2),
  P = c(P_demand, P_supply),
  Curve = factor(rep(c("Demand", "Supply"), each = length(Q_vals)))
)

cat("ANS of d.", "\n")
```

```
## ANS of d.
```

```
ggplot(df, aes(x = Q, y = P, color = Curve)) +
  geom_line(size = 1) +
  labs(title = "Supply and Demand Curves",
       x = "Quantity (Q)",
       y = "Price (P)") +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```


Supply and Demand Curves
Price (P)
80
60
40
10
15
20
25
Quantity (Q)
Curve
Demand
Supply